

• Moment generating function

$$M_X: \mathbb{R} \rightarrow [0, \infty)$$

expected value of an exponential function (e^{tx}) $\Rightarrow$ always positive

$$M_X(t) = E[e^{tx}]$$

$$M_X(t) = \sum_{x \in \Omega} e^{tx} \cdot p_X(x) \quad / \quad \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) \cdot dx$$

$$\text{Region of convergence: } D_X = \{t: M_X(t) < \infty\}$$

$t=0$ is always part of ROC

Differentiating the MGF and evaluating at origin gives moments.

e.g. MGF of Bernoulli r.v. $\begin{cases} 0 \text{ with probability } p \\ 1 \text{ with probability } (1-p) \end{cases}$

$$M_X(t) = E(e^{tx}) = e^{t \cdot p} + (1-p)$$

$$M_X'(t) = e^{t \cdot p} = p \text{ at } t=0 = \text{first moment}$$

e.g. exponential r.v.

$$\begin{aligned} M_X(t) &= E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} \cdot \lambda \cdot e^{-\lambda x} dx \\ &= \lambda \cdot \int_0^{\infty} e^{(t-\lambda)x} dx \\ &= \frac{\lambda}{\lambda - t}, \quad \lambda > t \end{aligned}$$

$$M_X: \mathbb{R} \rightarrow [0, \infty)$$

* If $M_X(t)$ is finite for all $|t| \leq \epsilon$ and for some $\epsilon > 0$, then $M_X(t)$ is infinitely differentiable on $(-\epsilon, \epsilon)$.

$$\text{Let } M_X^{(r)}(t) = \frac{d^r}{dt^r} M_X(t)$$

$$\text{Intuitively, } M_X^{(r)}(t) = E[X^r \cdot e^{tx}] \quad \forall r$$

$$\therefore E[X^r] = M_X^{(r)}(0)$$

• MGF of sums of independent r.v.s

$$Z = X + Y$$

$$M_Z(t) = E[e^{tZ}] = E[e^{t(X+Y)}] = E[e^{tX} \cdot e^{tY}]$$

$$= E[e^{tX}] \cdot E[e^{tY}]$$

since X and Y are independent

$$= M_X(t) \cdot M_Y(t)$$

* What about $M_Z(t)$ when $Z = X_1 + X_2 + \dots + X_N$ where N is a positive discrete random variable?

$$M_Z(t) = E[e^{tZ}] = E_N[E[e^{tZ}|N]] = E_N[(M_X(t))^N]$$

$$= \sum_n p_N(n) \cdot M_X(t)^n$$

$$\text{Q: Prove } M_Z(t) = M_N(\log M_X(t))$$

$$\text{Let } s = \log M_X(t)$$

$$M_N(s) = \sum_n p_N(n) \cdot M_X(s)^n$$

$$= \sum_n p_N(n) \cdot e^{sn}$$

$$= \sum_n M_Z(t)^n \cdot p_N(n)$$

$$= M_Z(t)$$

• STOCHASTIC SIMULATION

Suppose you have access to 5000 samples from a uniform random variable U over support $[0, 1]$.

We can use these 5000 samples and convert them into outcomes of a dice.

$$\begin{aligned} u < 1/6 &\longrightarrow 1 \\ 1/6 < u < 2/6 &\longrightarrow 2 \\ 2/6 < u < 3/6 &\longrightarrow 3 \\ 3/6 < u < 4/6 &\longrightarrow 4 \\ 4/6 < u < 5/6 &\longrightarrow 5 \\ 5/6 < u < 6/6 &\longrightarrow 6 \end{aligned}$$

• The Inverse Transform Method

Consider a discrete random variable X with support set $\{x_0, x_1, \dots\}$ and pmf $p_X(x_j) = p_j$ for $j = 0, 1, \dots$ such that $\sum_j p_j = 1$.

We wish to create i.i.d samples of X using i.i.d samples of $U[0, 1]$.

$$X = \begin{cases} x_0 & \text{if } u < p_0 \\ x_1 & \text{if } p_0 \leq u < p_0 + p_1 \\ x_2 & \text{if } p_0 + p_1 \leq u < p_0 + p_1 + p_2 \\ \vdots & \vdots \\ x_j & \text{if } \sum_{i=0}^{j-1} p_i \leq u < \sum_{i=0}^j p_i \\ \vdots & \vdots \end{cases}$$

A sample of X is generated using the sample of U as follows

$$X = x_j \text{ if } \sum_{i=0}^{j-1} p_i \leq U < \sum_{i=0}^j p_i$$

$$\Rightarrow F_X(x_{j-1}) \leq U < F_X(x_j)$$

After generating a random number U , we determine the value of X by finding the interval $[F_X(x_{j-1}), F_X(x_j))$ in which u lies.

$\therefore$ At a high level, we are performing $X = F_X^{-1}(U)$.